leftward). If one compresses the star, so its density increases and its circumference decreases (leftward motion in the drawing), then the star's pressure goes up in the manner of the solid curve, with a sharper rise at low densities where the resistance to compression is ⅔, and a slower rise at high densities where it is ⅓. This same compression of the star causes the star's surface to move in toward its center, thereby increasing the strength of the star's internal gravity in the manner of the dashed lines. The rate of gravity's increase is analogous to a ⅓ resistance: There is a ⅓ percent increase in gravity's strength for each 1 percent compression. The drawing shows several dashed gravity lines, one for each value of the star's mass, because the greater the star's mass, the stronger its gravity.

Inside each star, for example a 1.2-solar-mass star, gravity and pressure must balance each other. The star, therefore, must reside at the intersection of the dashed gravity line marked "1.2 solar masses" and the solid pressure curve; this intersection determines the star's circumference (marked on bottom of graph). If the circumference were larger, then the star's dashed gravity line would be above its solid pressure curve, gravity would overwhelm pressure, and the star would implode. If the circumference were smaller, pressure would overwhelm gravity, and the star would explode.

The intersections of the several dashed lines with the solid curve correspond to the masses and circumferences of equilibrium white dwarfs, as shown in the right half of Figure 4.3. For a star of small mass (lowest dashed line), the circumference at the intersection is large. For a star of higher mass (higher dashed lines), the circumference is smaller. For a star with mass above 1.4 Suns, there is no intersection whatsoever; the dashed gravity line is always above the solid pressure curve, so gravity always overwhelms pressure, no matter what the star's circumference may be, forcing the star to implode.